Computer Networks (CS-241)

Dr. Syed Muhammad Mohsin
Department of Computer Science,
University of Wah

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Computer Networking: A Top-Down Approach

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Types of Computer Networks (1/3)

PAN – Personal Area Network

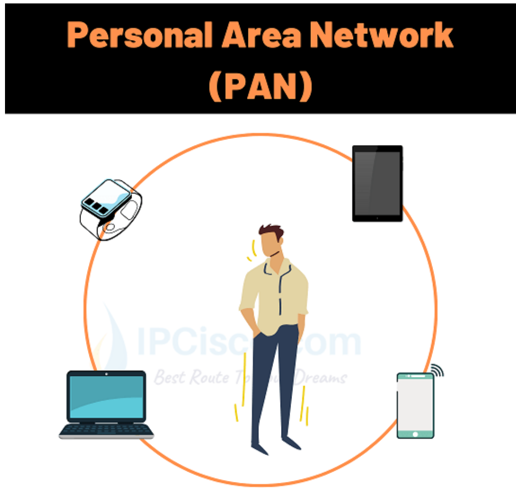
A PAN is the smallest type of network, designed for a single person's devices within a very small area (a few meters).

Key Features:

- Covers a few meters (like around a person's desk)
- Used for connecting personal devices such as laptops, smartphones, tablets, printers, etc.

Examples:

- Connecting a smartphone to a laptop via Bluetooth or hotspot.
- A wireless headset connected to a mobile phone.
- A USB-connected printer or mouse.



LAN – Local Area Network

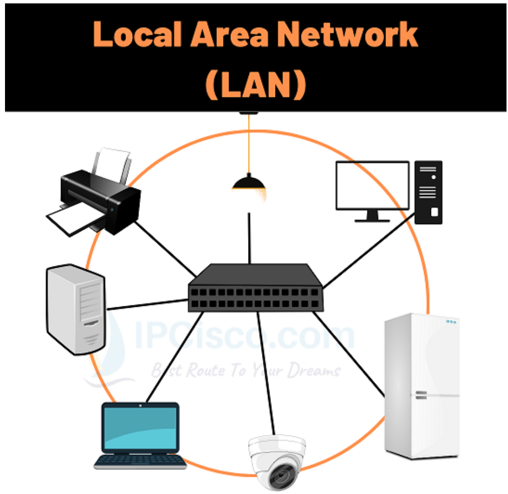
A LAN connects computers and devices in a limited geographical area such as a home, office, or building.

Key Features:

- High data transfer rates (up to 1 Gbps or more)
- Privately owned and managed
- Usually uses Ethernet cables or Wi-Fi

Examples:

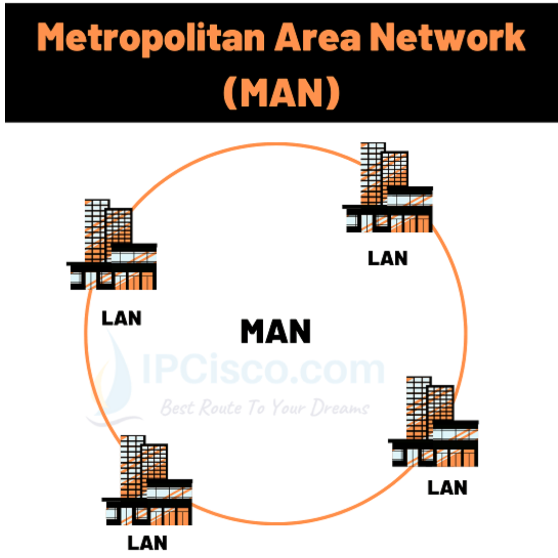
- Computers connected in a university lab or office.
- A Wi-Fi network in a home or school.
- A small company's internal network.



Types of Computer Networks (2/3)

MAN – Metropolitan Area Network

A MAN covers a larger area than a LAN — typically a city or campus — and interconnects multiple LANs.



Key Features:

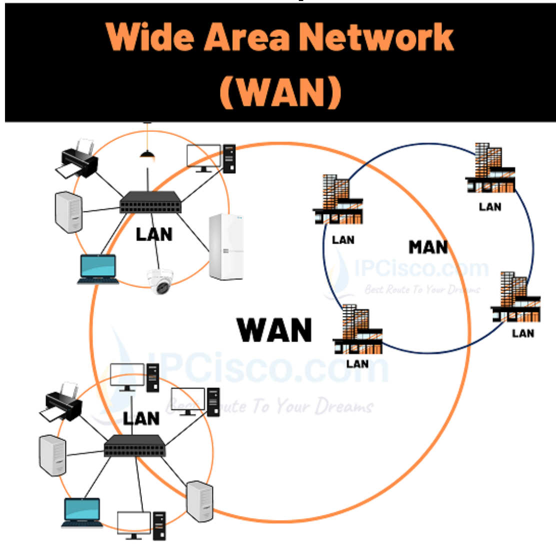
- Spans across a city or large campus
- Can be owned by an organization or managed by an Internet Service Provider (ISP)
- Uses high-speed fiber optic connections

Examples:

- A university's campus network connecting different buildings.
- City-wide Wi-Fi or cable TV networks.
- A government department's network across multiple city offices.

WAN – Wide Area Network

A WAN covers a very large geographical area — a country, continent, or even the whole world. It connects multiple MANs and LANs.



Key Features:

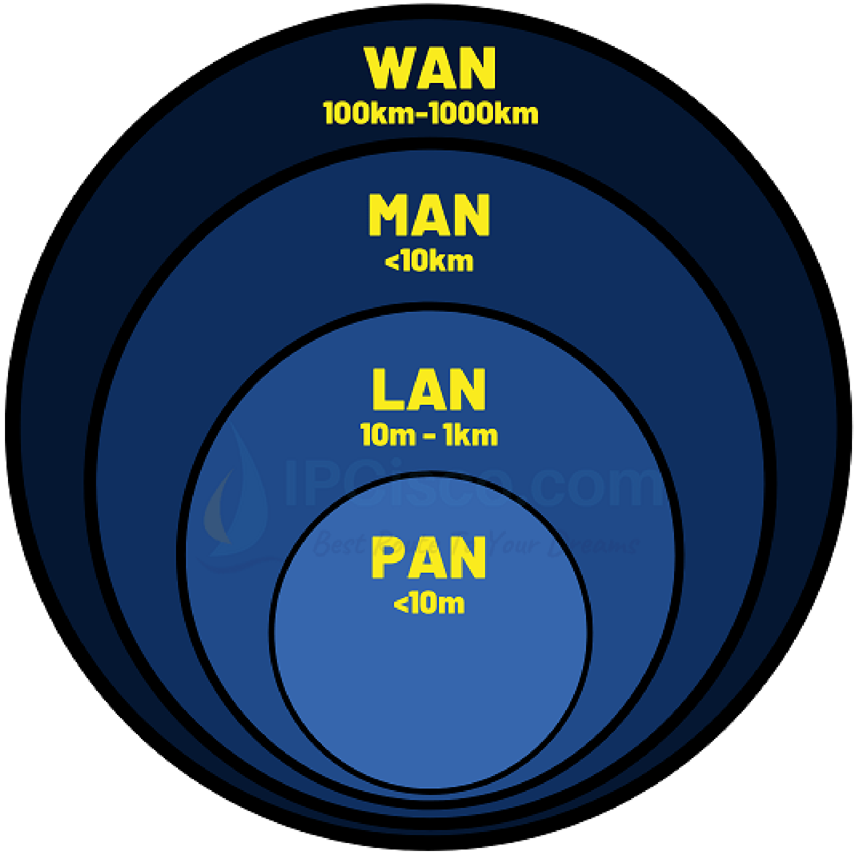
- Covers large distances
- Uses satellite links, undersea cables, or leased telephone lines
- Managed by multiple organizations or service providers

Examples:

- The **Internet** (largest WAN in the world).
- A multinational company's private network connecting offices in different countries.
- A bank's inter-branch communication network.

Types of Computer Networks (3/3)

Type	Full Form	Coverage Area	Ownership	Example
PAN	Personal Area Network	A few meters	Individual	Bluetooth between phone and laptop
LAN	Local Area Network	Building or campus	Private	Office network, home Wi-Fi
MAN	Metropolitan Area Network	City or campus	Organization/ISP	University or city-wide network
WAN	Wide Area Network	Country or global	Multiple organizations	Internet, corporate WAN



Types of Data Communications (1/4)

Data communication refers to the exchange of data between two devices through some transmission medium (like cables or wireless signals). The direction in which data can flow between sender and receiver defines the type of communication.

1. Simplex Communication (Sender → Receiver):

In Simplex, data flows only in one direction — from the sender to the receiver.
The receiver cannot send back any data or response.

Key Features:

- One-way communication
- Sender only transmits; receiver only receives
- Used where no feedback is required

Examples:

- Keyboard → CPU
- Monitor ← CPU (display output)
- Television broadcasting

Types of Data Communications (2/4)

2. Half Duplex Communication (Sender ⇄ Receiver (but one at a time)):

In Half Duplex, data flows in both directions, but only one direction at a time. The sender and receiver take turns to send or receive data.

Key Features:

- Two-way communication, but not simultaneous
- One device must wait while the other transmits
- Reduces collisions but is slower than full duplex

Examples:

- Walkie-talkies (one speaks while the other listens)
- Police radio systems
- Traditional CB* radio communication

*CB (Citizens Band) Radio is a two-way radio communication system that allows individuals to communicate over short distances using radio frequency channels without needing a license.

Types of Data Communications (3/4)

3. Full Duplex Communication (Sender ⇄ Receiver (simultaneous)):

In Full Duplex, data flows in both directions simultaneously. Both devices can send and receive data at the same time.

Key Features:

- Two-way communication at the same time
- More efficient and faster than half duplex
- Requires two separate communication channels (one for each direction)

Examples:

- Telephone conversations
- Video calls
- Modern computer networks (Ethernet, Wi-Fi)

Types of Data Communications (4/4)

Type	Direction of Data Flow	Simultaneous Communication	Examples
Simplex	One-way only	No	Keyboard to CPU, Monitor display
Half Duplex	Both ways (one at a time)	No	Walkie-talkie, CB radio
Full Duplex	Both ways (simultaneous)	Yes	Telephone, Internet communication

Network Topologies (1/6)

A network topology refers to the physical or logical layout of how computers, cables, and other network devices are connected in a network. It shows how data flows between nodes (devices).

1. Bus Topology

All computers and devices are connected to a **single central cable** called the **bus** or **backbone**.

Advantages:

- Easy to install and inexpensive (uses less cable)
- Works well for small networks

Disadvantages:

- If the main cable fails, the entire network stops working
- Data collisions can occur when two devices send data at the same time
- Difficult to troubleshoot

Examples:

- Early Ethernet networks
- Small temporary networks

Network Topologies (2/6)

2. Star Topology

All devices are connected to a **central hub or switch**. The hub acts as a repeater to transmit data between devices.

Advantages:

- Easy to install and manage
- Failure of one cable doesn’t affect others
- Easy to add or remove devices

Disadvantages:

- If the central hub/switch fails, the whole network goes down
- Requires more cable length than bus topology

Examples:

- Modern Ethernet networks in offices or homes
- Wi-Fi networks using a router as a central point

Network Topologies (3/6)

3. Ring Topology

Each device is connected to **two other devices**, forming a **closed loop (ring)**. Data travels in **one direction** (or both in dual ring).

Advantages:

- Data flows in an orderly manner (less collision)
- Performs better under heavy load than bus topology

Disadvantages:

- If one device or cable fails, the whole network may stop
- Difficult to troubleshoot and modify

Examples:

- Token Ring networks (IBM)
- Some fiber-based MANs

Network Topologies (4/6)

4. Tree Topology (Hierarchical Topology)

A combination of **Star** and **Bus** topologies. Devices are arranged in a **hierarchical structure** — like branches of a tree — with a **root node** and sub-nodes.

Advantages:

- Scalable and easy to expand
- Easy to manage and detect faults
- Supports large networks

Disadvantages:

- If the main hub fails, the entire section goes down
- Requires a lot of cables and hardware

Examples:

- Corporate networks (universities, enterprises)
- Used in Wide Area Networks (WANs)

Network Topologies (5/6)

5. Mesh Topology

Every device is connected to **every other device** directly. There are two types:

- **Full Mesh:** All devices interconnect directly.
- **Partial Mesh:** Only some devices are interconnected.

Advantages:

- Very reliable — no single point of failure
- Data can take multiple paths to reach destination
- High performance and security

Disadvantages:

- Expensive due to large cabling and configuration
- Complex to manage

Examples:

- Internet backbone connections
- Military or mission-critical networks

Network Topologies (6/6)

Topology	Layout Description	Advantages	Disadvantages	Example
Bus	Single backbone cable	Cheap, simple	Cable failure affects all	Early LANs
Star	All devices connect to hub	Easy to manage	Hub failure affects all	Modern LANs
Ring	Devices in a closed loop	Orderly data flow	One failure breaks network	Token Ring
Tree	Hierarchical structure	Scalable	Hub failure affects section	University networks
Mesh	Devices interconnect	Reliable, fast	Expensive, complex	Internet backbone